

CSE 3172: Mobile Application Development Lab

Credits	1
Contact Hours	28
Year	Third
Semester	First
Prerequisite	CSE 1222: Object Oriented Programming Lab, CSE 2232: User Interface Development Lab
Course Type	☐ Theory ☒ Laboratory work ☐ Project work ☐ Viva Voce

Motivation

Mobile applications are a primary mode of software use today. This course introduces React Native so students can extend their React.js knowledge to build cross-platform mobile apps with native UI, device APIs, and mobile deployment workflows.

Course Objective

This laboratory course introduces cross-platform mobile application development using React Native and Expo. Students will build multi-screen apps, use device APIs, manage local data, and produce a distributable build using EAS.

Course Outcomes (COs), Program Outcomes (POs) and Assessment

CO No.	CO Statement	Corresponding PO	Domain / Level of Learning Taxonomy	Delivery Methods & Activities	Assessment Tools
CO1	To set up React Native / Expo projects and mobile UIs using core components and Flexbox layouts	Modern Tool Usage (PO5)	Psychomotor domain – Level 3	☐ Lecture Note ☒ Text Book ☐ Audio/Video ☒ Web Material ☒ Lab Manual	☒ CA ☒ Final Exam ☐ Assignment ☐ Notebook ☐ Presentation
CO2	To construct multi-screen mobile applications using React Navigation and device APIs	Modern Tool Usage (PO5)	Psychomotor domain – Level 4	☐ Lecture Note ☒ Text Book ☐ Audio/Video ☒ Web Material ☒ Lab Manual	☒ CA ☒ Final Exam ☒ Assignment ☐ Notebook ☐ Presentation
CO3	To develop a data-driven React Native application using the Fetch API and EAS Build	Design/Development of Solutions (PO3)	Cognitive domain – Level 6	☐ Lecture Note ☒ Text Book ☐ Audio/Video ☒ Web Material ☒ Lab Manual	☒ CA ☒ Final Exam ☐ Assignment ☐ Notebook ☒ Presentation

Assessment and Marks Distribution

- Continuous Assessments (CA) (30%)
- A comprehensive final exam + Lab note book (60%)
- A class participation mark (10%).

Lab Course Contents / List of Experiments

- Lab 1 — Environment Setup and React Native Fundamentals
 - Lab 2 — Core Components, Styling, and Flexbox
 - Lab 3 — Navigation with React Navigation
 - Lab 4 — User Input, Forms, and Keyboard Handling
 - Lab 5 — Fetching Data and Local Storage
 - Lab 6 — Device APIs: Camera, Location, and Notifications
 - Lab 7 — EAS Build: Producing a Distributable App
 - Lab 8 — React Native Capstone Project
-

Textbooks

1. Nader Dabit — *React Native in Action*, Manning Publications, 2019
2. Bonnie Eisenman — *Learning React Native* (2nd ed.), O'Reilly Media, 2018